

BIOPROCESS ENGINEERING

EFFECTS OF NONDIMENSIONALIZATION ON PHYSICS-INFORMED NEURAL NETWORK (PINN) APPLIED TO A LACTIC ACID PRODUCTION BIOREACTOR SIMULATION

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INTRODUCTION

NN (Neural Network) is a class of highly flexible algorithms capable of working as universal approximators of any given function with inputs and outputs. [1] PINN (Physics-Informed Neural Networks) is a framework launched in 2018 that supports the optimization of NNs using a small amount of data (small data) [2]. This work studies nondimensionalization strategies (NdS) applied to solve partial differential equations (PDEs) that represent the production of lactic acid by *L. casei* in a batch bioreactor using the PINN framework.

METHODOLOGY

A batch bioreactor for the production of lactic acid (LA) by *Lactobacillus casei* using lactose as a substrate was simulated using PINN with soft constraints on the programming language Python and the library DeepXDE. [3] Obtained PINN models have one input variable, *t* (time), and four output variables: *X* (biomass concentration), *P* (lactic acid concentration), *S* (lactose concentration) and *V* (reactor liquid volume). The error was calculated using Mean Absolute Deviation (MAD) of each output variable, comparing PINN and a reference numerical method (finite differences).

RESULTS AND DISCUSSION

Figure 1 shows the values of MAD for each NdS applied to the simulated system. The nondimensional "x10" and "d10" mean a scaling effect was applied to the output of the PINN, with "x10" meaning scaling factor was multiplied by 10 and "d10" divided by 10. The common nondimensionalization produce average MAD of 12.54, while the dimensional version had a 2.92 MAD. The scaling x10 produced the highest error and the d10 scaling the lowest, with respective average MAD of 21.95 and 1.81.

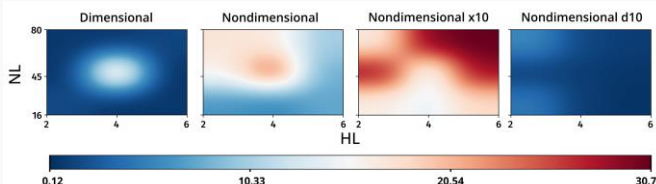


Figure 1: MAD vs NL (neurons per layer) and HL (number of hidden layers).

Figure 2 displays the output variables (*X*, *P*, *S* and *V*) for a PINN with NL=16, HL=6 without any NdS. PINN, Num (numerical method result) and Experimental (experimental data from source [4]) are displayed. The training region is of 10 h, only half of the simulation time.

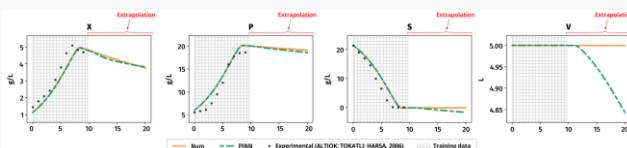


Figure 2: Output variables for a dimensional PINN with NL=16, HL=6.

Figure 3 displays the output for a PINN with NL=16, HL=6 and Nondimensionalization d10. The model displayed good capabilities of extrapolation, respected physical and biological constraints and reproduced almost perfectly the data inside the training region. The only significant deviation was found on biomass concentration, after 15h, and was still relatively small.

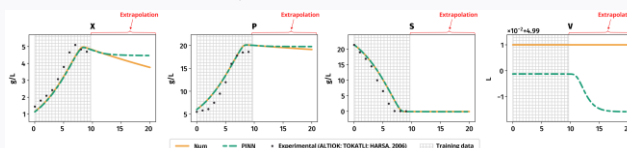


Figure 3: Output variables for a PINN with NL=16, HL=6 and Nondimensionalization d10.

CONCLUSIONS

The nondimensionalization of output variables with d10 (scaling factor divided by 10) produced the best results, with an average MAD of 1.81. NdS were capable of improve the results obtained and allowed more simple (lower values of NL and HL) PINNs to produce simulations with lower errors in comparison to the dimensional PINNs. This can allow PINNs to be used more frequently in bioprocess simulation and also in realtime bioprocess simulation, which can increase both economical productivity and process security.

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